

CareerGhana Python Internship — Task 1

Deliverable: Interactive Career Path Quiz CLI Tool & Documentation
Repository: https://github.com/J-git22/careerghana_tasks/tree/main/task_1_career_quiz

1. Project Overview & Objective

The Career Quiz CLI tool is an interactive terminal application designed to guide aspiring technology professionals toward their optimal career specialization. By answering 7 targeted questions exploring developer problem-solving style, preferred work environments, and core interests, users receive an evidence-backed recommendation across 5 key industry tracks:

- **Software Engineering / Full-Stack:** Building resilient end-to-end applications and backend architectures.
- **Data Science & Machine Learning:** Extracting insights from data, statistics, and predictive modeling.
- **Cybersecurity Analyst:** Auditing vulnerability vectors, threat analysis, and infrastructure defense.
- **DevOps & Cloud Engineering:** Server automation, cloud provisioning, CI/CD, and site reliability.
- **Product Design & UI/UX:** Crafting human-centered user experiences, design systems, and visual interfaces.

2. Career Track Evaluation Framework

Track ID	Career Pathway	Core Evaluation Focus
SE	Software Engineering	Algorithms, OOP, system architecture, debugging code
DS	Data Science / AI	Statistics, data modeling, exploratory analysis, metrics
CS	Cybersecurity	Network defenses, penetration testing, cryptography, audit logs
DO	DevOps & Cloud	Containers, Linux shell, infrastructure-as-code, reliability
UI	Product / UI/UX Design	Design empathy, user flow, prototyping, accessibility

3. Algorithm & Scoring Logic

Each of the 7 multiple-choice questions maps candidate responses to categorical weight vectors. At completion, scores are normalized into percentage affinity scores, highlighting primary and secondary affinities. The tool outputs visual progress bars, detailed descriptions, curated learning roadmaps, and immediate actionable steps.

4. Sample Terminal Execution Output

```
=====
■ YOUR CAREER RECOMMENDATION ■
=====
Top Match: SOFTWARE ENGINEERING / FULL-STACK DEVELOPMENT (85.7%)
Affinity Breakdown:
[1] Software Engineering | [#####] 85.7%
[2] DevOps & Cloud | [##### ] 57.1%
[3] Data Science / ML | [##### ] 28.6%
=====
```

5. Technical Highlights & Standards

- **Zero External Dependencies:** Built strictly using Python 3 standard library modules for universal compatibility.
- **Defensive Input Validation:** Robustly handles invalid inputs, case variations, whitespace, EOF, and keyboard interrupts.
- **Cross-Platform:** Clean execution across Windows, macOS, and Linux terminals.